MICHIGAN MEDICINE
UNIVERSITY OF MICHIGAN

PEDIATRIC PATIENTS ≤50 kg PRE-OP AND INTRAOPERATIVE ANTIBIOTIC DOSING RECOMMENDATIONS[ξ,Σ]

Antibiotic Dose	Recommended Concentration	Infusion Time	Patient Weight in kg												Intraoperative Redosing Interval
			< 5	5-7.49	7.5-9.9	10-14.9	15-19.9	20-24.9	25-29.9	30-34.9	35-39.9	40-44.9	45-50	> 50	
Ampicillin 50 mg/kg (Ampicillin/Sulbactam dosed on ampicillin) max: 2000 mg	1 g/10 mL	Maximum of 200 mg/min	USE MG/KG DOSING	375 mg	500 mg	750 mg	1000 mg	1250 mg	1500 mg	1750 mg	2000 mg	2000 mg	2000 mg	REFER TO >50 kg DOSING RECOMMENDATIONS	*q2h x2 redoses*
Aztreonam 30 mg/kg max: 2000 mg	1 g/10 mL	IVP 3-5 minutes		225 mg	300 mg	450 mg	600 mg	750 mg	750 mg	1000 mg	1000 mg	1000 mg	1500 mg		*q4h x2 redoses*
Cefazolin (non-cardiac/redoses) 30 mg/kg max: 2000 mg	1 g/10 mL	IVP 3-5 minutes		225 mg	300 mg	450 mg	600 mg	750 mg	750 mg	1000 mg	1000 mg	1000 mg	1500 mg		*q4h x2 redoses*
*Cefazolin (cardiac/SBE**) 50 mg/kg max: 2000 mg	1 g/10 mL	IVP 3-5 minutes		375 mg	500 mg	750 mg	1000 mg	1250 mg	1500 mg	1750 mg	2000 mg	2000 mg	2000 mg		*q4h x2 redoses w/non-cardiac redose value* ***(30 mg/kg)***
Cefepime 50 mg/kg max: 2000 mg	1 g/10 mL	3-5 minutes		375 mg	500 mg	750 mg	1000 mg	1250 mg	1500 mg	1750 mg	2000 mg	2000 mg	2000 mg		*q4h x2 redoses*
Cefotaxime 50 mg/kg max: 2000 mg	1 g/10 mL	3-5 minutes		375 mg	500 mg	750 mg	1000 mg	1250 mg	1500 mg	1750 mg	2000 mg	2000 mg	2000 mg		*q3h x2 redoses*
Cefoxitin 40 mg/kg max: 2000 mg	1 g/10 mL	3-5 minutes		300 mg	400 mg	600 mg	800 mg	1000 mg	1250 mg	1500 mg	1500 mg	2000 mg	2000 mg		*q2 x2 redoses*
Ceftriaxone 50 mg/kg max: 2000 mg	40 mg/mL	3-5 minutes		375 mg	500 mg	750 mg	1000 mg	1250 mg	1500 mg	1750 mg	2000 mg	2000 mg	2000 mg		*None*
Cefuroxime 50 mg/kg max: 1500 mg	1 g/10 mL	3-5 minutes		375 mg	500 mg	750 mg	1000 mg	1500 mg	1500 mg	1500 mg	1500 mg	1500 mg	1500 mg		*q4h x2 redoses*
Ciprofloxacin 10 mg/kg max: 400 mg	2 mg/mL	Minimum of 60 minutes		75 mg	100 mg	150 mg	200 mg	250 mg	300 mg	350 mg	400 mg	400 mg	400 mg		*None*
Clindamycin (non-cardiac/redoses) 10 mg/kg max: 900 mg	20 mg/mL	Maximum of 30 mg/min		75 mg	100 mg	150 mg	200 mg	250 mg	300 mg	350 mg	400 mg	450 mg	500 mg	REFER	*q6h x2 redoses*

ANTIMICROBIAL STEWARDSHIP
MICHIGAN MEDICINE

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			< 5	5-7.49	7.5-9.9	10-14.9	15-19.9	20-24.9	25-29.9	30-34.9	35-39.9	40-44.9	45-50	> 50	
*Clindamycin (cardiac/SBE** in PCN allergy) 20 mg/kg max: 900 mg	20 mg/mL	Maximum of 30 mg/min	USE MG/KG DOSING	150 mg	200 mg	300 mg	400 mg	500 mg	600 mg	700 mg	800 mg	900 mg	900 mg		*Q6h x2 redoses w/non-cardiac redose value* ***(10 mg/kg)***
Co-trimoxazole 5 mg/kg TMP Not in infants <2 months max: 160 mg TMP	80 mg SMX: 16 mg TMP/mL 5 ml vial Max conc 1:10	Minimum of 30 minutes		37.5 mg	50 mg	75 mg	100 mg	125 mg	150 mg	160 mg	160 mg	160 mg	160 mg		*Q6h x2*
Fluconazole 6 mg/kg max: 400 mg	2 mg/mL	Maximum of 200 mg/hour		40 mg	50 mg	75 mg	100 mg	135 mg	165 mg	200 mg	225 mg	250 mg	285 mg		*None*
Gentamicin/ Tobramycin 5 mg/kg max: 250 mg	10 mg/mL	Minimum of 30 minutes		USE MG/KG DOSING If patient is receiving systemic aminoglycoside therapy prior to surgery, consult OR pharmacist to determine appropriate timing and necessity of peri-operative dosing											*None*
Levofloxacin 10 mg/kg max: 750 mg	5 mg/mL	Minimum of 60 minutes		75 mg	100 mg	150 mg	200 mg	250 mg	300 mg	350 mg	400 mg	450 mg	500 mg		*None*
Metronidazole 15 mg/kg (dosing rounded to be divisible by 5) max: 500 mg	5 mg/mL	Minimum of 30 minutes		100 mg	150 mg	200 mg	300 mg	375 mg	450 mg	500 mg	500 mg	500 mg	500 mg		*None*
Piperacillin/ Tazobactam 100 mg/kg based on Piperacillin max: 3000 mg	100 mg/mL	Minimum of 30 minutes		750 mg	1000 mg	1500 mg	2000 mg	2500 mg	3000 mg	3000 mg	3000 mg	3000 mg	3000 mg		*Q2h x2 redoses*
Vancomycin (cardiac) 10 mg/kg max: 1000 mg	1 g/100 mL	Minimum of 60 minutes		USE MG/KG DOSING If patient is receiving systemic vancomycin therapy prior to surgery, consult OR pharmacist to determine appropriate timing and necessity of peri-operative dosing											*Q12h x2 redoses*

I. INTRODUCTION

The use of peri-operative antimicrobials has become an essential component of the standard of care for certain surgical procedures and can result in a reduced risk of post-operative infection when sound and appropriate principles are utilized. However, the benefit of antimicrobial prophylaxis must be weighed against the risks of toxic and allergic reactions, emergence of resistant bacteria, drug interactions, super-infection, and cost.

II. CONSIDERATIONS FOR ANTIMICROBIAL PROPHYLAXIS

Goal: Administer antimicrobial prophylaxis to achieve serum and tissue levels of antimicrobial at the time of incision and for the duration of the operation, that are in excess of the minimum inhibitory concentration (MIC) needed for organisms that may be encountered during the procedure.

a. Antimicrobial prophylaxis should be administered if there is a risk of infection in the absence of a prophylactic agent; clean procedures rarely require prophylaxis unless high risk procedure, including implantation of prosthetic material.
    i. Clean procedures are defined as those with no acute inflammation or transection of gastrointestinal, oropharyngeal, genitourinary, biliary, or respiratory tracts (elective cases, no technique break).
b. The activity of the chosen prophylactic agent(s) should encompass the most common pathogens associated with the surgical procedure and consider local susceptibility data, but need not cover every likely pathogen.
c. The prophylactic agent must be administered in a dose which provides an effective tissue concentration prior to incision / intra-operative bacterial contamination.
    i. In most instances, a single intravenous dose of an antimicrobial agent provides adequate tissue concentrations around the time of anesthesia induction and throughout the operation.
        1. Antimicrobial agent infusion should begin 15-60 minutes before the incision with the exception of vancomycin, levofloxacin, ciprofloxacin, gentamicin, azithromycin and fluconazole. These infusions should begin 45-90 minutes before the incision and infused over 60-120 minutes as indicated (See tables above**).**
        2. **All prophylactic antimicrobials should be discontinued after the intra-operative period, unless otherwise specified.**
            a. Data have not supported subsequent doses after surgical closure and may increase the risk of *Clostridium difficile* and antimicrobial resistance.
            b. A longer duration of antimicrobials may be indicated, if concomitant infection is present at the time of surgery.

Low-risk allergies include: pruritus without rash, remote (>10 years) unknown reaction, patient denies allergy but is on record, mild/severe rash with no other symptoms (see exceptions in 'severe
non-IgE reactions below), urticaria/hives with no other symptomsSee β-lactam allergy evaluation and empiric guidance for further information.

[3] High-risk allergies include: respiratory symptoms, angioedema, cardiovascular symptomssyncope/passing out, arrhythmia), anaphylaxis. **If a patient has a high-risk allergy to penicillins, they may receive cefazolin, cefoxitin, or cefuroxime.**. See β-lactam allergy evaluation and empiric guidance for further information.

[4] Severe non-IgE reactions that are contraindications to further beta-lactam use (**except aztreonam, which can be used unless the reaction was to ceftazidime or cefiderocol**) unless approved by Allergy: organ damage (kidney, liver), drug-induced immune-mediated anemia/thrombocytopenia/leukopenia, rash with mucosal lesions (Stevens Johnson Syndrome/Toxic Epidermal Necrosis), rash with pustules (acute generalized exanthematous pustulosis), rash with eosinophils and organ injury (DRESS – drug rash eosinophilia and systemic symptoms), rash with joint pain, fever, and myalgia (Serum Sickness). See β-lactam allergy evaluation and empiric guidance for further information.

[5] Beta-lactam antibiotics include the following antibiotic classes: penicillins, cephalosporins, carbapenems, aztreonam

α Refer to Pre-op and Intraoperative Antibiotic Dosing Recommendations Guidelines

- Guideline includes infusion duration and time to intra-operative redosing
- All prophylactic antimicrobials should be discontinued after the intra-operative period, unless otherwise specified
- Pediatrics patients > 50 kg should receive antibiotic dosing based on > 50 kg guideline
- Use mg/kg dosing for patients < 5 kg
- SIP operative pediatrics procedures include: cardiac, ventriculoperitoneal shunts, and spinal surgery

* Redosing should occur with the same initial dose, with the exception of cefazolin and clindamycin when used in cardiac/SBE prophylaxis

** SBE prophylaxis, ACC/AHA 2008 guideline update on valvular heart disease

π Adapted from Clinical Infectious Diseases 2004;38:1706-15 and Am J Health-Syst Pharm 2013;70.

γ Reconstituted dose injected directly into vein or via running IV fluids (only if IV piggyback not available).

δ Intermittent IV infusion.

ε Gentamicin dose should be based on ideal body weight unless actual body weight is less than ideal body weight. Use alternative to gentamicin in if SCr ≥ 2 mg/dL or CrCL < 40 mL/min, unless otherwise specified.

ζ Daptomycin dosing is based on actual body weight

λ Infusion Timing:

- Infusions should begin 15-60 minutes prior to incision for all antimicrobial agents unless listed below:
  - Levofloxacin, ciprofloxacin, vancomycin, gentamicin, azithromycin, and fluconazole, which should begin 45-90 minutes prior to incision.
- If pre-operative antibiotics have already been administered but incision has been delayed more than 60 minutes beyond the maximum dosing window (i.e., beta-lactam antibiotics started >2 hours prior to incision), consider pre-operative re-dosing for all antibiotics except vancomycin and aminoglycosides. For vancomycin and aminoglycosides, please contact pharmacy to determine if re-dosing is appropriate.

ξ Patients receiving systemic antibiotics prior to procedure should still receive the standard pre-operative antimicrobial prophylaxis with appropriate timing of administration as outlined in the UMHS surgical antimicrobial prophylaxis guidelines. Given the risk of nephrotoxicity, in patients receiving vancomycin or aminoglycosides prior to procedure who need these agents for pre-operative antimicrobial prophylaxis, please consult pharmacy to see if treatment doses can be rescheduled such that administration begins 45-90 minutes prior to incision.

Σ In patients with known colonization or infection with drug-resistant pathogens, the standard pre-operative antimicrobial prophylaxis should generally still be administered unless otherwise specified in procedure-specific guidelines. Please contact the ID approval pager (36149) to discuss the case if concerned about: history of a multidrug-resistant organism (e.g., ESBL- producing organism, carbapenem-resistant Enterobacteriaceae, etc.) at the site of the procedure in the previous year OR history of MRSA colonization in the previous year.

Ω Deviations may be appropriate for staged procedures. For example, in TMJ replacement cases ampicillin/sulbactam prior to the intra-oral portion of the procedure followed by cefazolin prior to the neck incision.

Last revised:	Antimicrobial Stewardship Subcommittee:	P&T:

References

	Citation
1	Milstone A. et al. Timing of Preoperative Antibiotic Prophylaxis: A modifiable risk factor for deep surgical site infections after pediatric spinal fusion. Pediatr Infect Dis J. 2008 Aug;27(8):704-8.
2	Kids' Campaign 2007 Pediatric Webcast Series: Reduce Surgical Complications and Prevent Surgical Site Infections. 5 Million lives Campaign/
3	Bratzler DW., Houch PM. Antimicrobial Prophylaxis for Surgery: An Advisory Statement from the National Surgical Infection Prevention Project. Am J Surg. 2005 Apr;189(4):395-404.
4	American Academy of Pediatrics: Antimicrobial Prophylaxis in Pediatric Surgical Patients. Pediatrics. 1984 Sep;74(3):437-9.
5	Bratzler DW, et al. Clinical practice guidelines for antimicrobial prophylaxis in surgery. Am J Health Syst Pharm. 2013 Feb 1;70(3):195-283.

Revision History

Date	Revision
08/2020	Revised OB/GYN section
10/2020	Revised Urology section
11/2020	Revised Gastrointestinal section
03/2021	Revised allergy wording, Orthopedic, Plastics, and Cardiothoracic sections
04/2021	Adjusted rhinoplasty/septorhinoplasty and vasectomy recommendations
12/2021	Revised Orthopedic section
03/2022	Revised dosing recommendations for patients ≤ 50 kg
10/2022	Added Radiology, updated allergy wording
11/2022	Revised pediatric vancomycin, gentamicin/tobramycin dosing
01/2023	Adjusted allergy wording, revised beta-lactam footnote
03/2023	Removed redosing recommendation for piperacillin-tazobactam during liver transplant
08/2023	Revised Gastrointestinal section
02/2024	Revised Cardiothoracic section
12/2024	Revised Orthopedic section for open fractures
03/2025	Replaced vancomycin with daptomycin throughout document; revised daptomycin dosing

The recommendations in this guide are meant to serve as treatment guidelines for use at Michigan Medicine facilities. If you are an individual experiencing a medical emergency, call 911 immediately. These guidelines should not replace a provider's professional medical advice based on clinical judgment, or be used in lieu of an Infectious Diseases consultation when necessary. As a result of ongoing research, practice guidelines may from time to time change. The authors of these guidelines have made all attempts to ensure the accuracy based on current information, however, due to ongoing research, users of these guidelines are strongly encouraged to confirm the information contained within them through an independent source.